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ORIGINAL ARTICLE

Investigation of sustainable face milling of austenitic stainless steel: material removal performance and tool modification

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Abstract
This study investigates face milling of austenitic stainless steel under sustainable dry and Minimum Quantity Lubrication (MQL) conditions, with material removal volume (MRV) as the primary response variable. Austenitic stainless steel poses significant machining challenges due to its low thermal conductivity, high ductility, and work hardening tendencies, exacerbating tool wear mechanisms like notching and chipping. A full factorial design of experiments was conducted, varying cutting speed and feed per tooth, to evaluate their effects on MRV. Quadratic regression models were developed, incorporating Box-Cox transformations for improved robustness and predictive accuracy. Monte Carlo simulations were employed to assess robustness and process variability. Results show that dry machining yields a higher MRV (optimal: 250 cm³ at 213 m/min, 0.20 mm/tooth), while MQL provides a higher material removal rate but lower MRV (optimal: 231 cm³ at 170 m/min, 0.32 mm/tooth), illustrating the trade-off between tool life and short-term productivity. Experimental validation with new test conditions confirmed the accuracy and robustness of the regression models, with predictions closely matching observed results within a 95% prediction interval. Monte Carlo simulations further highlighted that dry machining resulted in higher MRV with greater variability, while MQL gave more consistent but lower MRV values. Observations of notch wear and cracks (most plausibly associated with thermal fatigue) under MQL motivated a preliminary tool modification that redirected the MQL jet away from the cutting edge during tool exit, showing up to 25–50% improvement in MRV. These findings contribute to sustainable machining strategies by providing actionable insights into parameter selection for improved efficiency and durability in industrial applications.

Keywords Milling · Austenitic stainless steel · Tool wear · Notch wear · Thermal fatigue · Monte Carlo · MQL · Tool design

1 Introduction

Austenitic stainless steel is widely used in the chemical processing, automotive (mainly for molds of plastic components), and aerospace industries due to its exceptional corrosion resistance and mechanical properties. However, its high ductility, low thermal conductivity, and tendency to work harden create significant challenges during machining [1]. Poor thermal conductivity of this material, measured at 16.2 W/mK, coupled with intermittent machining applications, can induce thermal cracks on tool coatings [2]; the main wear mechanism is often adhesive, but combined with other degradations. Recent advancements in cutting tool coatings, such as TiAlSiN, have shown promise, particularly under Minimum Quantity Lubrication (MQL) conditions, where friction and cutting forces are reduced compared to dry machining. However, deposits on the rake (built-up edge formation) and thermal fatigue remain critical issues under dry machining, necessitating further exploration of optimized cutting parameters [2].

Machining in austenitic stainless steel is prone to notching, adversely affecting edge stability, tool life, and surface integrity. Notching at the depth of cut occurs due to the rubbing interaction between the machined surface and the

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cutting tool. This phenomenon is driven by a combination of wear mechanisms, including abrasion, adhesion, surface fatigue, and tribochemical reactions [3], which lead to localized and strong adhesive interactions between the chip and the tool material [4]. Milling operations exacerbate the notching problem as cracks further compromise the cutting edge, accelerating wear progression. Thermal and mechanical cracking, particularly during high-speed milling, results from cyclic loading and contributes to cutting edge degradation. These mechanisms, combined with high stresses and temperature variations at the depth of cut, significantly increase wear rates and reduce tool life [5]. While the notch wear can sometimes be mitigated by altering the tool path [6] or adjusting the depth of cut [7], these strategies are not universally applicable and may be constrained by the specific milling application.

Recent studies have shown that deformation-hardened layers, workpiece inclusions such as SiO_2, and high contact pressures are critical factors in the formation of early notch wear during machining [8–10]. Furthermore, optimized MQL and hybrid cooling strategies have been reported to mitigate the severity of notch wear and thermal fatigue effects, especially in difficult-to-machine stainless steels [11, 12].

The interplay of cutting parameters profoundly impacts machining performance and tool wear. Studies like Malekan et al. [5] have demonstrated that during end milling AISI 316 L, higher feed rates enhance material removal rates (MRR) at the expense of accelerated tool wear. For turning AISI 304, Binali et al. [13] found that higher cutting speeds improve surface finish under MQL conditions. These findings underscore the necessity of optimizing cutting parameters to balance productivity and tool longevity, particularly under varying lubrication conditions. Studies on machining austenitic grades like AISI 316 have emphasized the importance of balancing cutting data to optimize MRR and surface quality. For example, Equbal et al. [14] demonstrated that depth of cut significantly influences MRR, while feed rate primarily impacts surface roughness during slot milling, underscoring the need for precise control of machining parameters. These insights are crucial for addressing the challenges associated with machining similar materials.

Most studies on milling processes have shown the impact on end milling applications, and few are related to face milling with indexable inserts. Selaimia et al. [15] investigated dry face milling of X2CrNi18-9 (AISI 304 L) using low axial depth of cut, a_p = 0.3 to 0.9 mm, and feed per tooth, f_z = 0.08 to 0.25 mm/tooth. Their study identified feed per tooth as the most influential parameter for MRR. However, the study's focus on dry machining with low depths of cut and feed limits its applicability to industrial production settings, where higher depths of cut and feed are typically used. This highlights the need for research on face milling under more practical conditions, exploring broader feed ranges, higher depth of cut, and comparative lubrication strategies such as dry and MQL machining to better represent real-world applications.

Recent studies have aimed to enhance the machinability of stainless steel through innovative tool designs and sustainable approaches. In [16], the sustainable face milling characteristics of AISI 304 using serrated cutting edges were investigated. The results showed reductions in cutting power and specific cutting energy by 8–17% compared to conventional cutting edges, highlighting the potential of innovative tool geometries to enhance efficiency in machining conditions. However, the study primarily focused on cutting forces and power consumption without addressing tool wear, MRV, or MRR, leaving gaps in understanding the broader performance implications of such tool designs under varying lubrication conditions.

While traditional cooling and lubrication methods using flood coolant effectively control tool wear and heat generation, they pose significant environmental and health concerns. Korkmaz et al. [17] emphasize that such practices contribute to waste generation, carbon emissions, and adverse health effects due to exposure to petroleum-based cutting fluids. These challenges necessitate a shift toward sustainable manufacturing strategies, such as dry and near dry, as MQL offers reduced environmental impact. MQL offers a sustainable alternative that reduces friction and improves cooling efficiency with minimal fluid use. However, understanding the trade-offs between tool wear mechanisms and machining efficiency under MQL and dry conditions remains critical for optimizing the milling of stainless steel. Research has extensively explored the relationship between cutting parameters and machining performance. While MQL has shown a potential to reduce friction at the tool-chip interface and improve surface quality, intermittent cooling under MQL may exacerbate thermal fatigue, particularly in ductile materials like austenitic stainless steel. TiAlSiN coatings have been shown to improve tool life and surface integrity under MQL conditions compared to traditional coatings [2]. However, challenges such as adhesive wear and built-up edge formation persist, especially in dry machining environments.

Dry machining eliminates cooling-related concerns but struggles with reducing adhesive wear and mitigating work hardening effects. Comparative studies of MQL versus dry milling under equivalent experimental conditions for austenitic stainless steel are sparse, limiting the ability to assess their relative merits systematically. Jeulin et al. [18] demonstrated that tool wear can account for up to 90% of the environmental impact under specific cutting conditions, emphasizing the critical role of cutting speed and tool life in balancing productivity and sustainability. Their study also highlighted the utility of Monte Carlo simulations in analyzing uncertainties

and validating machining models, providing a robust framework for evaluating process outcomes. This underscores the importance of incorporating environmental considerations and advanced modeling techniques into machining research.

Previous research has highlighted the significant influence of variations in workpiece microstructure and composition, particularly oxide inclusions, on tool wear progression [19]. To address this variability, our study ensures that all workpieces are sourced from the same material batch, minimizing compositional differences and enabling a more robust evaluation of machining performance under different lubrication conditions. Additionally, the combined effects of cutting speed and feed per tooth on MRV and tool wear remain poorly understood, particularly when considering critical mechanisms like thermal fatigue and notching.

Advanced statistical methods, such as Box-Cox transformations and Monte Carlo simulations, provide an opportunity to develop a more robust framework for improving process outcomes. For example, Nagendra et al. [20] utilized the Box-Cox transformation in turning to optimize process parameters for surface roughness, demonstrating its effectiveness in fine-tuning models. Adopting such techniques in face milling can enhance the predictive accuracy of MRV and tool wear.

This study investigates face milling of austenitic stainless steel under sustainable dry and MQL conditions as material removal volume as response variable. Cutting speed and feed per tooth were selected as the main variables, while depth of cut and milling width were fixed at values to isolate their influence on the dominant tool failure mechanisms of notch wear. This focused design enables robust statistical modeling without diluting experimental power. The novelty of this work lies in (i) addressing the underexplored case of face milling with indexable inserts under sustainable lubrication strategies, (ii) applying Box–Cox regression modeling combined with Monte Carlo simulations to quantify robustness against process variability, and (iii) introducing and preliminarily validating a modified tool concept for MQL jet shielding that shows potential for extending tool life. Together, these contributions provide actionable insights into sustainable machining strategies, balancing productivity and durability in industrial applications.

2 Experimental procedures

Square bars with dimensions 40 mm × 40 mm and a length of 200 mm were used as workpiece material during the milling tests. According to the supplier certificate, the material is classified as EN 1.4301 (AISI 304), although the chemical composition with a low carbon content (C = 0.018%) also meets the criteria for EN 1.4307 (AISI 304 L). This stainless steel offers excellent corrosion resistance, high strength, and good ductility. It is widely used in industrial applications such as chemical processing equipment, structural components, and various types of machinery. The matrix material consists of austenite with a Brinell hardness of approximately 200 HB, which can increase due to work hardening. The certified chemical composition of the alloy is given in Table 1.

Face milling was conducted in a vertical machine, DMU 40 DECKEL MAHG, using a Sandvik Coromant R245-12 T3 K-MM S30T cutter with PVD-coated inserts (TiAlN). The experiments employed a full factorial design with three levels of cutting speed, v_c: 170, 210, 250 m/min, and feed per tooth, f_z: 0.20, 0.26, 0.32 mm. Axial depth of cut (a_p = 1.25 mm) and radial depth of cut (a_e = 40 mm) were selected to ensure stable tool engagement. This choice avoids foot forming [21] and prevents the generation of thin chips at tool entry and exit, thereby promoting smooth engagement and disengagement of the cutter and eliminating wear mechanisms associated with these phenomena. The experiments were performed with a single cutting insert mounted in the milling cutter to minimize the potential influence of the runout of the cutting edges during the trials. The experimental setup is shown in Fig. 1.

Together with the two categorical conditions, dry and MQL, a total of 36 experiments were conducted in random order. Each of the 18 unique parameter combinations (3 cutting speeds × 3 feed per tooth levels × 2 lubrication

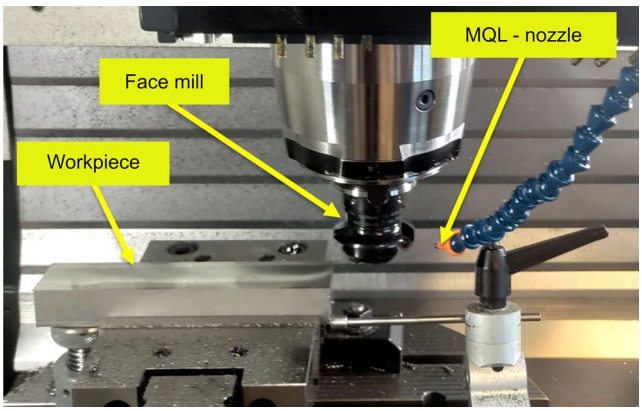


Fig. 1 Photo of the experimental setup

Table 1 Main chemical composition of the workpiece material, from supplier certificate [wt%]

C	Si	Mn	Cr	Ni	*P*	S	*N*
0.018	0.46	1.74	18.29	8.05	0.029	0.028	0.093

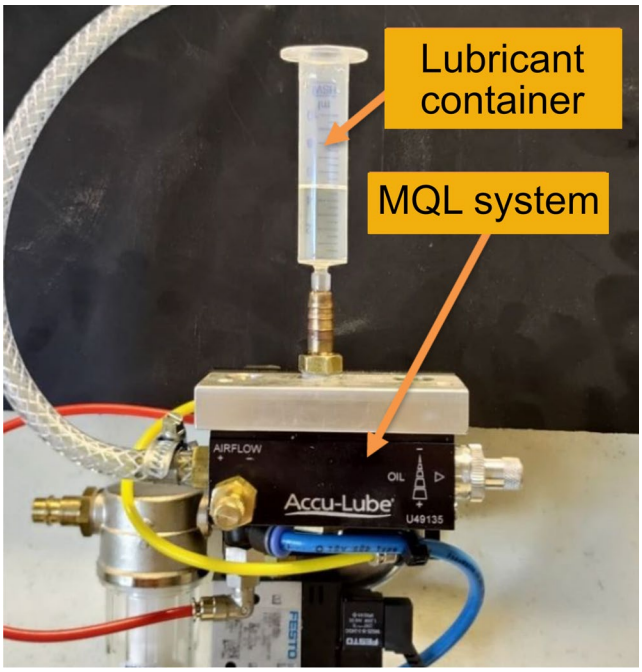


Fig. 2 The MQL system. The lubricant container is designed to store and supply the lubricant in precise, controlled quantities to the dosing pump

strategies) was repeated once, resulting in one replicate per condition. These replicates were used to estimate random error. In addition, six independent validation tests were later conducted, which were not part of the factorial design but were used to evaluate the predictive capability of the regression models.

The MQL parameters, such as nozzle position, angle, and distance to the cutting edge, were carefully adjusted in collaboration with the supplier to ensure oil mist application at the correct location and timing. While the lubricant flow was fixed, the experimental focus was on achieving the proper jet direction rather than varying velocity or flow rate, as previous experience had indicated that the location and timing of oil delivery were the most critical factors for reducing adhesion-related wear. ECCOLUBRIC E200L was delivered at 8 ml/h using a modified setup, ensuring controlled lubricant application (see Fig. 2). The nozzle was positioned at approximately 8 cm from the tool tip and oriented at an angle of about 45° relative to the spindle axis. The air pressure was maintained at 10 bar. This configuration ensured that the oil mist was directed toward the cutting zone at tool entry. In dry machining, no lubrication was applied. Material removal volume was recorded as the primary response variable.

The milling operation was interrupted at defined intervals to inspect tool wear, with the inspection frequency adjusted according to the cutting conditions and observed wear. Preliminary tests indicated that up to eight passes could be performed before the first inspection without exceeding the wear criteria. Subsequent inspections were then carried out at shorter intervals, typically between one and six passes, depending on the progression of tool wear. Each cutting pass corresponded to a machining length of 200 mm and 10 cm^3 of removed material. The tool was examined under a general-purpose digital microscope, and the response variable was recorded once the notch depth reached the defined tool wear criteria.

Surface roughness measurements were conducted using a Mitutoyo SJ-210 stylus profilometer at the end of the machining trials.

Tool wear in metal cutting is complex, but typical wear observed in this work on the tool faces and the cutting edge after face milling austenitic stainless steel is shown in Fig. 3. No significant flank wear was observed.

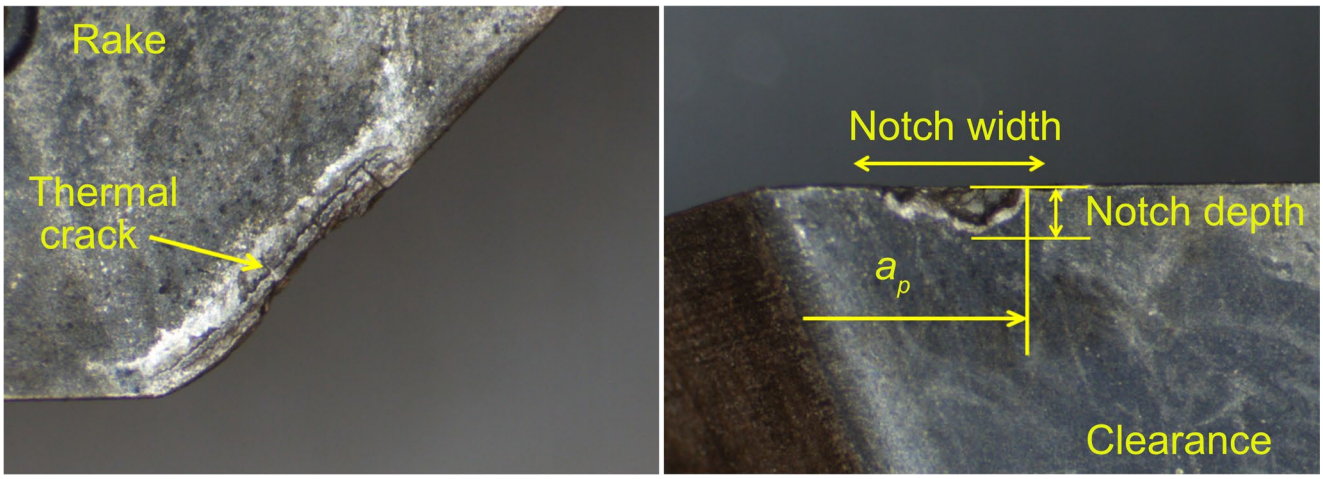


Fig. 3 Typical wear observed on the tool faces and cutting edge after milling in austenitic stainless steel. These include notching and thermal cracks. Notching is the critical indicator in this work. Depth of cut denoted as a_p

3 Results and discussions

The results of the experimental design matrix with corresponding material removal volumes are summarized in Table 2. For each of the 18 parameter combinations, both the main trial and one replicate are reported as individual values rather than averages, making the natural variability visible and providing the basis for estimating random error. The data also indicates a nonlinear relationship between the control factors and the response, which forms the foundation for the subsequent statistical analysis and optimization.

A quadratic regression model was built with only second-order interaction for each categorical factor. The normality of residuals for dry and MQL conditions was evaluated using graphical methods (histograms and Q-Q plots) and statistical tests (Lilliefors and Anderson-Darling). Although normality was supported, a Box-Cox transformation was applied to stabilize the variance further and improve model robustness. The estimated transformation parameter (likelihood estimator) λ are $\lambda_{Dry} = 0.0051$ for dry and $\lambda_{MQL} = 0.0511$ for MQL conditions. The transformed response, MRV_{TF}, was used in subsequent regression analysis:

$$MRV_{TF} = \frac{MRV^{\lambda} - 1}{\lambda} \tag{1}$$

To interpret the results in the original scale, the transformed material removal volume, MRV_{TF} models were back-transformed to their original form using the inverse Box-Cox transformation:

$$MRV = (\lambda \cdot MRV_{TF} + 1)^{1/\lambda} \tag{2}$$

Table 2 Design matrix with response

Exp. no	Control factor			Response	
	Categorical	Continuous		Main trails	Replicates
	Conditions	v_c [m/min]	f_z [mm]	MRV [cm^3]	MRV [cm^3]
1	Dry	250	0.32	160	140
2	Dry	250	0.26	180	170
3	Dry	250	0.20	200	200
4	Dry	210	0.32	200	160
5	Dry	210	0.26	290	230
6	Dry	210	0.20	240	300
7	Dry	170	0.32	230	250
8	Dry	170	0.26	210	240
9	Dry	170	0.20	220	160
10	MQL	250	0.32	120	120
11	MQL	250	0.26	150	180
12	MQL	250	0.20	200	190
13	MQL	210	0.32	190	160
14	MQL	210	0.26	240	190
15	MQL	210	0.20	140	150
16	MQL	170	0.32	260	200
17	MQL	170	0.26	160	190
18	MQL	170	0.20	130	110

3.1 Regression analysis and model adequacy

The regression coefficients and their corresponding standard error and test statistics are shown in Tables 3 and 4. For dry conditions (Table 3), the regression model is simplified by the elimination of the feed per tooth squared term. The intercept was retained despite its non-significant p-value to avoid imposing an unnecessary constraint forcing the model through the origin. Since the model is valid for $v_c \in [170, 250]$ m/min and $f_z \in [0.2, 0.32]$ mm/tooth, excluding the intercept is unjustified and could reduce predictive accuracy within the design space. The remaining terms are all statically significant because their p-values are less than the significance level of 0.05. For MQL conditions (Table 4), the cutting speed squared term is eliminated due to a high p-value.

The goodness of fit is summarized as follows—the models display satisfactory adjusted coefficients of determination. For dry conditions, the adjusted R-squared = 0.509, indicating that the model explains about 51% of the variability. For MQL conditions, the adjusted R-squared = 0.723, indicating that the model explains about 72% of the variability. The standard error of the regression Root Mean Squared Error, RMSE, is 0.153 for dry conditions and 0.164 for MQL conditions.

Model validation was done by analysis of variance. The ANOVA results (This distinction between Lack of Fit and Pure Error ensures that the observed residual variability is

Table 3 Estimated coefficients and statistics for MRV, dry conditions

Term	Coefficient	Standard error	t-value	*p*-value
Constant	−1.7191	2.4197	−0.71048	0.48996
v_c [m/min]	0.060057	0.020975	2.8632	0.013315
f_z [mm]	10.742	4.7963	2.2397	0.04322
v_c [m/min] f_z [mm]	−0.05727	0.022568	−2.5376	0.024765
v_c [m/min] v_c [m/min]	−0.00011424	0.000048	−2.3861	0.032929

Table 4 Estimated coefficients and statistics for MRV, MQL conditions

Term	Coefficient	Standard error	t-value	*p*-value
Constant	−5.8502	2.0129	−2.9064	0.012257
v_c [m/min]	0.038526	0.0063926	6.0266	0.000042
f_z [mm]	59.482	12.911	4.6072	0.000491
v_c [m/min] f_z [mm]	−0.15266	0.024162	−6.318	0.000027
f_z [mm] f_z [mm]	−50.331	22.78	−2.2095	0.045696

due to random noise rather than deficiencies in the model structure. The non-significant Lack of Fit p-values further reinforces the suitability of the reduced quadratic models in capturing the relationship between input factors and material removal volume.

Tables 5 and 6 show that the models are statistically significant with a p-value of 0.008663 for dry condition and 0.000257 for MQL condition, indicating the predictors explain the variation in machined volume very well. The Lack of fit p values are greater than 0.05, indicating no significant lack of fit, which confirms the reduced models are appropriate. This distinction between Lack of Fit and Pure Error ensures that the observed residual variability is due to random noise rather than deficiencies in the model structure. The non-significant Lack of Fit p-values further reinforces the suitability of the reduced quadratic models in capturing the relationship between input factors and material removal volume.

The transformed model equations on material removal volume under dry and MQL conditions are:

$$MRV_{TF,\ Dry} = -1.7191 + 0.0601\ v_c + 10.7420\ f_z - 0.0573\ v_c f_z - 0.0001\ v_c^2 \quad (3)$$

$$MRV_{TF,\ MQL} = -5.8502 + 0.0385\ v_c + 59.4820\ f_z - 0.1527\ v_c f_z - 50.3310\ f_z^2 \quad (4)$$

Back-transformed models in the original scales become

$$MRV_{Dry} = \left(0.0051 \cdot \left(-1.7191 + 0.0601\ v_c + 10.7420\ f_z - 0.0573\ v_c f_z - 0.0001\ v_c^2\right) + 1\right)^{1/0.0051} \quad (5)$$

$$MRV_{MQL} = \left(0.0511 \cdot \left(-5.8502 + 0.0385\ v_c + 59.4820\ f_z - 0.1527\ v_c f_z - 50.3310\ f_z^2\right) + 1\right)^{1/0.0511} \quad (6)$$

These back-transformed models enable direct interpretation of the predicted material removal volume in cubic centimeters for the specific ranges of cutting speed and feed per tooth.

Table 5 Analysis of variance for the regression model on MRV, dry conditions

Source	Sum of squares	Degree of freedom	Mean squares	F-value	*p*-value
Total	0.81298	17	0.047823		
Model	0.50787	4	0.12697	5.4098	0.008663
Residual	0.30511	13	0.02347		
Lack of fit	0.14634	4	0.036585	2.0738	0.167060
Pure error	0.15877	9	0.017641		

Table 6 Analysis of variance for the regression model on MRV, MQL conditions

Source	Sum of squares	Degree of freedom	Mean squares	F-value	*p*-value
Total	1.6486	17	0.096977		
Model	1.2989	4	0.32472	12.071	0.000257
Residual	0.34971	13	0.026901		
Lack of fit	0.13563	4	0.033908	1.4255	0.301640
Pure error	0.21408	9	0.023787		

3.2 Response contour and optimization

Figure 4 shows the contours of MRV by Eqs. (5–6). During dry conditions, moderate cutting speeds and low feed per tooth reduce notching and result in high MRV values. During MQL conditions, high feed per tooth and low cutting speed or high cutting speed in combination with low feed per tooth provide high MRV. Generally, dry conditions provide higher MRV.

To identify cutting parameters that maximize predicted MRV within the experimental domain, constrained nonlinear optimization was performed separately for dry and MQL conditions using the reduced quadratic regression models

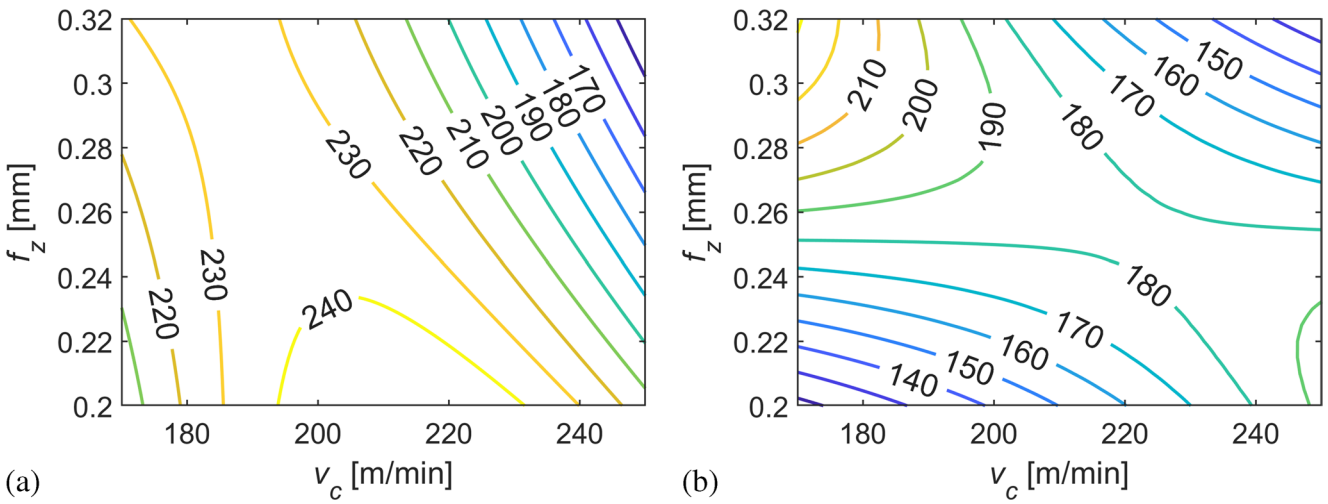


Fig. 4 Contour plot of material removal volume in cubic centimeters versus cutting speed v_c and feed per tooth f_z. (**a**) Dry conditions, and (**b**) MQL conditions

described in Sect. 3.1. Because the Box–Cox back-transformation is monotonic for $\lambda > 0$, maximizing the transformed prediction corresponds directly to maximizing MRV in the original scale. The following explicit optimization problem was formulated:

$$\max_{v_c,\, f_z} \widehat{\mathrm{MRV}}\,(v_c, f_z) \text{ such that } \begin{cases} v_c \in [170,\ 250]\ \mathrm{m/min}, \\ f_z \in [0.20,\ 0.32]\ \mathrm{mm/tooth}, \\ (\alpha_p = 1.25\ \mathrm{mm},\ \alpha_e = 40\ \mathrm{mm}) \end{cases} \tag{7a}$$

Or equivalently:

$$\min_{v_c,\, f_z} -\widehat{\mathrm{MRV}}_{\mathrm{TF}}\,(v_c, f_z) \text{ such that } \begin{cases} v_c \in [170,\ 250]\ \mathrm{m/min}, \\ f_z \in [0.20,\ 0.32]\ \mathrm{mm/tooth}, \\ (\alpha_p = 1.25\ \mathrm{mm},\ \alpha_e = 40\ \mathrm{mm}) \end{cases} \tag{7b}$$

No additional constraints were imposed; feasibility with respect to tool and machine limits was ensured by the experimental design. Interior-point algorithm via MATLAB® fmincon was used as the solution method, with the objective defined as in Eq. 7b. Multiple initial guesses (e.g., [170, 0.20], [200, 0.26], [250, 0.32]) were tested, and all converged to the same stationary solutions, confirming robustness with respect to starting conditions. The predicted MRV was then back-transformed using λ_{Dry} and λ_{MQL}. The identified optimal cutting parameters are:

- Dry conditions:

  Cutting speed v_c = 213 m/min,
  Feed per tooth f_z = 0.2 mm/tooth,
  The predicted material removal volume MRV at these parameters is 250 cm³.

- MQL conditions:

  Cutting speed v_c = 170 m/min,
  Feed per tooth f_z = 0.32 mm/tooth,
  The predicted material removal volume MRV at these parameters is 231 cm³.

The optimization (well in line with the trends observed in the response contours (Fig. 4)) revealed distinct cutting parameter combinations for maximizing material removal volume under dry and MQL conditions. The higher optimal feed rate, and lower cutting speed, in MQL are attributed to improved lubrication but constrained by thermal fatigue. Dry conditions, on the other hand, provide greater stability and MRV at lower feed rates and moderate cutting speeds.

For comparison, the corresponding material removal rates (MRR) at these optimized conditions were calculated using the standard MRR formula: 0.32 cm^3/min for dry machining and 0.41 cm^3/min for MQL. These calculated values were used only as a secondary indicator of productivity in the subsequent discussion. The optimization revealed distinct cutting parameter combinations for maximizing MRV under dry and MQL conditions. While MQL yields a higher material removal rate (MRR), it results in a lower MRV before reaching the tool wear limit. This reflects the classical trade-off between tool life and production rate. Dry machining extends tool life by enabling each tool to remove more material before replacement, whereas MQL, with its higher MRR, offers greater short-term throughput in continuous production where tool changes are minimized. However, MQL may exacerbate thermal fatigue due to intermittent cooling and potentially accelerating crack initiation, chipping, and edge failure at high cutting speeds in austenitic stainless steel. The regression models are calibrated for $v_c \in$ [170, 250] m/min and $f_z \in$ [0.20, 0.32] mm/tooth with a_p = 1.25 mm and a_e = 40 mm using the specified tool, machine, and lubrication setup. While the coefficients are specific to this configuration, the modeling and optimization workflow is directly transferable to other systems.

3.3 Monte Carlo simulations for robustness

Monte Carlo simulations based on the regression models were performed to assess robustness using factor tolerances and model noise. Simulations for the optimal parameters with 10 000 iterations were conducted, introducing variations of ±5% for cutting speed, ±1% for feed per tooth, and residual model noise (RMSE). For dry conditions, the results indicate: Mean volume: 249.27 cm³, Standard deviation: 37.42 cm³, 95% Confidence Interval: [174.43, 324.11] cm³. For MQL conditions, the simulation results indicate: Mean volume: 232.82 cm³, Standard deviation: 32.97 cm³, 95% Confidence Interval: [161.88, 298.76] cm³. The simulations suggest robust predictions within reasonable tolerances and provide insight into the variability of the machined volume.

Figure 5 provides a visual representation of the distribution of MRVs based on the Monte Carlo iterations. The histograms illustrate the spread of predicted MRV values, while the fitted normal distribution highlights the central tendency and variability for both dry and MQL conditions. The symmetry and central alignment of the distributions around the mean demonstrate the robustness of the model predictions. The relatively narrow spread (standard deviation, σ) suggests small sensitivity to factor and model noise. Recent studies have shown that combining physics-based models with AI-driven hybrid approaches can further enhance the predictive accuracy of tool wear monitoring [22]. Integrating such hybrid modelling with response surface methodology and Monte Carlo simulations could further strengthen

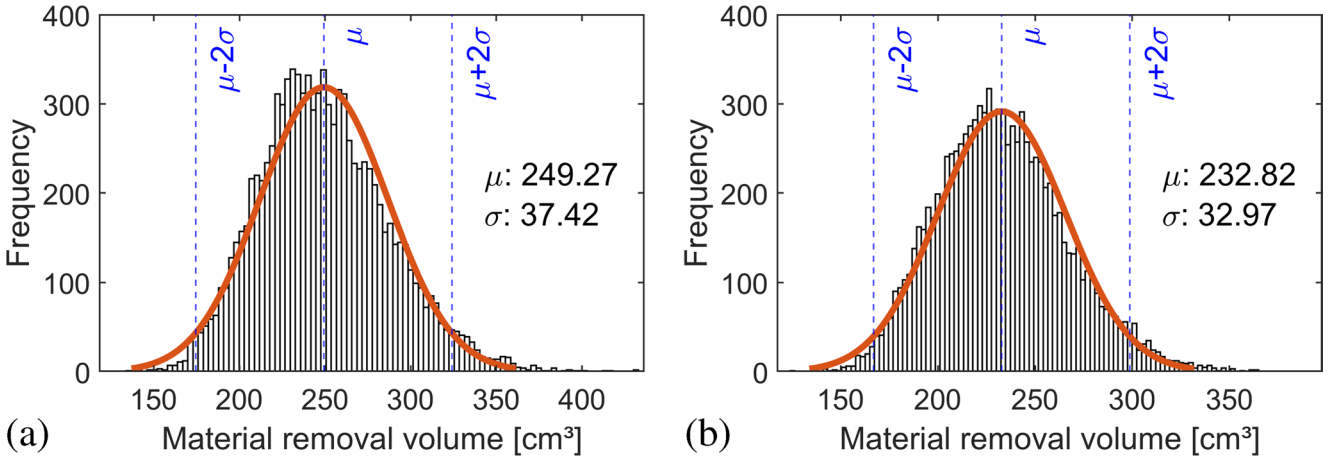


Fig. 5 Monte Carlo simulation for optimal parameters of material removal volume models (fitted normal distribution – curve). The simulation includes factor noise, and model noise. Summary statistics: mean μ, and standard deviation σ. Normal distribution is fitted to the data. (**a**) dry conditions, and (**b**) MQL conditions

future control and wear prediction. The fitted normal distributions align well with the calculated 95% confidence intervals. The visual representation highlights that the majority of predicted MRV values fall within these intervals, underscoring the reliability of the models. Comparing the distributions for dry (Fig. 5a) and MQL (Fig. 5b) conditions reveals that dry machining results in a higher mean MRV and exhibits slightly greater variability compared to MQL. These differences align with the distinct optimization results for the two conditions.

3.4 Experimental validation

Six additional tests without replication were conducted. The new observations were to check the predictive performance of the regression models when applied to data that were not used in the model estimation. Table 7 shows the experimental conditions and results for the new observations. Also, the predicted response using the regression model, together with the 95% prediction intervals are displayed.

The validation results show that the model provides acceptable predictions, with minor overestimations in some cases.

3.5 Improvement of the cutting tool and lubrication effects

In addition to notch wear at the depth of cut, small cracks were consistently observed on the cutting edges, most plausibly caused by thermal fatigue. While these cracks may remain harmless under stable conditions, their interaction with adhesion and cyclic mechanical loading can accelerate notch formation and eventually lead to edge chipping. The intermittent cooling associated with MQL application likely intensified this effect, as thermal shock occurs when the jet impinges on the edge immediately after disengagement. These observations motivated the proposed tool modification, where the MQL jet was redirected to avoid direct impingement during tool exit and instead applied primarily at entry, when the cutting edge is at its lowest temperature. Surface roughness was not significantly affected by tool wear, since the wiper flat exhibited only minor wear of about 20 µm, whereas notch wear at the main edge determined the tool life criteria. Chip formation observations supported these findings. Early in tool life, chips were tightly curled and uniform, indicating stable plastic deformation and low friction at the tool–chip interface. Later in the tool life, chips

Table 7 Experimental conditions and results for new observations together with calculated predicted values and 95% prediction interval for the measured response

Exp. No	Control factor			Measured Response	Predicted Response	Prediction interval	
						Lower	Upper
	Cindition	v_c [m/min]	f_z [mm]	MRV [cm^3]	MRV$_p$ [cm^3]	PI$_L$ [cm^3]	PI$_U$ [cm^3]
1	Dry	190	0.22	240	235.12	165.93	332.96
2	Dry	230	0.29	200	195.26	138.21	275.68
3	Dry	205	0.24	270	238.59	168.34	337.94
4	MQL	190	0.22	150	153.37	114.22	205.04
5	MQL	230	0.29	210	166.55	124.04	222.64
6	MQL	205	0.24	150	175.49	130.89	234.27



Fig. 6 Representative chip morphology during MQL milling at $f_z = 0.32$ mm and $v_c = 210$ m/min. The left chip was formed with a fresh, sharp cutting edge. The right chip was generated after partial edge rounding

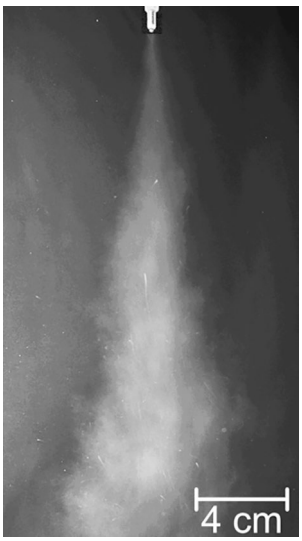


Fig. 7 Visualization of the MQL jet used in the experiments, illustrating the spray pattern and spatial extent of the lubricant flow

became more open and conical, reflecting increased edge rounding, higher friction, and elevated thermal loads (Fig. 6).

The existing MQL jet configuration is shown in Fig. 7. To realize this concept, a Polylactic Acid (PLA) plug was 3D printed and mounted inside the tool body (Fig. 8) to block direct airflow to the cutting edge during exit. Preliminary tests at $f_z = 0.32$ mm and varying cutting speeds showed notable improvements in tool life, with increases in MRV ranging from 25% to 50%. These results highlight the potential of MQL jet shielding as a practical and straightforward enhancement to mitigate thermal shock and extend tool performance.

These observations support the hypothesis that effective MQL lubrication reduces the adhesive wear mechanism, leading to lower cutting forces and temperatures. Therefore, proper lubrication in the cutting zone leads to thinner chips with shorter contact lengths, which tend to curl more. As the temperature decreases, the abrasive wear mechanism is also suppressed, further reducing abrasive wear. A reduction in cutting forces under MQL conditions has similarly been reported in turning operations [23]. Consequently, under efficient lubrication conditions, a higher feed per tooth, f_z, can be applied, improving overall productivity while maintaining acceptable tool wear.

4 Conclusions

The reduced and Box-Cox transformed quadratic model provides a robust and statistically significant regression model for MRV under dry and MQL conditions.

1. Dry machining offers a higher MRV and probably better tool life than MQL in milling austenitic stainless steel when using the settings and our MQL system, and the high cutting speed range, above the Build up Edge speeds.
2. Optimal cutting data for dry conditions were $v_c = 213$ m/min and $f_z = 0.20$ mm, achieving an MRV of 250 cm³.

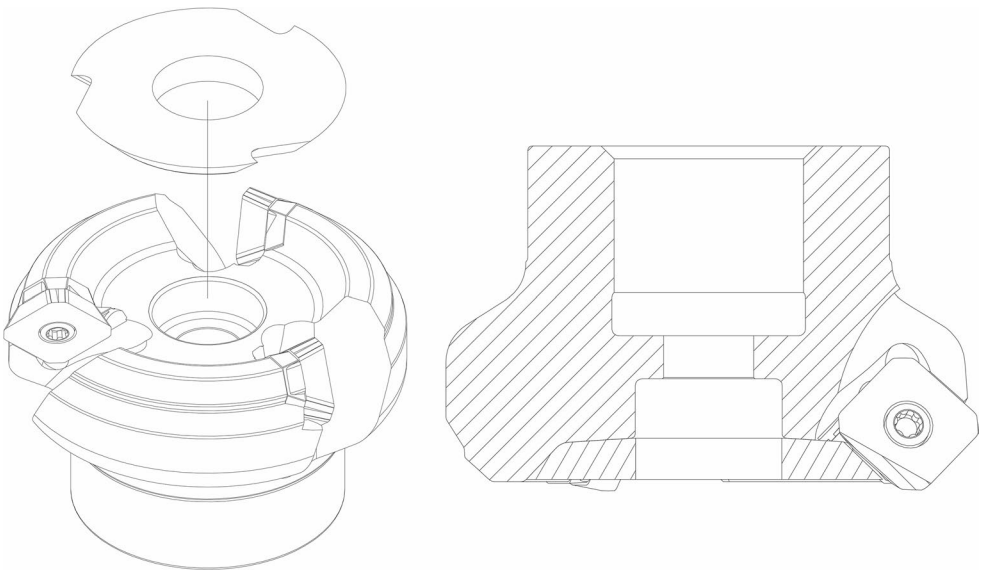


Fig. 8 Modified milling tool (R245-050Q22-12 L) with an internally mounted plug designed to block the direct MQL jet as the cutter disengages the workpiece, thereby reducing thermal shock to the cutting edge

3. Optimal cutting data for MQL conditions were $v_c = 170$ m/min and $f_z = 0.32$ mm, achieving an MRV of 231 cm³. Despite the lower MRV, MQL yielded higher MRR, highlighting the trade-off between tool life and short-term production rate. Shorter tool life under MQL conditions is likely due to the intensified effects of thermal fatigue.
4. The Monte Carlo simulation results provide practical insight into the expected variability of MRV. Such information allows manufacturers to account for tolerances during process planning and ensures robustness in achieving target MRV values under dry and MQL conditions. They also show that dry machining gives higher MRV with greater variability, while MQL yields lower MRV with more consistent performance.
5. Preliminary results from a modified milling tool, designed to shield the cutting edge from direct MQL jet exposure during tool exit, indicated up to 50% improvement in MRV, suggesting potential for extending tool life.

Enhanced MQL efficiency can reduce cutting forces and wear, enabling higher feeds and increased productivity. Future work should explore alternative MQL configurations and lubrication strategies to enhance lubrication efficiency and mitigate thermal fatigue, including jet application techniques that minimize airflow on the cutting edge immediately after disengagement, when thermal loads are at their peak. Additionally, integrating hybrid modeling approaches could further enhance tool wear monitoring and process control.

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Data availability Not applicable.

Code availability Not applicable.

Declarations

Conflict of interest The authors declare no conflict of interest.

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